

The Watermark Theorem

The discriminant of the line lattice of Fermat surfaces of prime degree

Rafael Amichis Luengo · Madrid · tretoef@gmail.com · github.com/tretoef-estrella

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Version note (30 September 2026). The statement and the proof are unchanged. Three additions, after reading the sources in the original:

1. Correction. §1 says that Shioda «asked for its structure» in 1987. He asked for the determinant only: [7], p. 133, Question 7.2 (characteristic $p \equiv 1 \pmod m$) and Question 7.4, formula (7.10) (the complex surface), both for prime m . The structure of the discriminant group, which goes beyond his question, is the companion *Double Ladder Theorem* (same repository).

2. Literature. In 2015 Shioda still used the formula as expected, not as proved: it appears inside Conjecture 23 of T. Shioda, *Mordell–Weil lattice of higher genus fibration on a Fermat surface*, J. Math. Sci. Univ. Tokyo **22** (2015), 443–468, §6.5. [6, Table 1] lists the elementary divisors for every $m \leq 14$; for $m = 5, 7, 11, 13$ their product is $m^{3(m-3)^2}$.

3. Verified in Lean. On 30 September 2026 the theorem was certified in Lean 4 with Mathlib, along a second route that is entirely integral and uses no characters. For every prime $m \geq 5$, Lean proves that V is free of rank $3(m-1)(m-2)+1$ and that the Gram determinant of every basis of V equals $m^{3(m-3)^2}$, with sign $+$. The proof depends only on the three standard axioms of Lean. That the Gram matrix is the intersection matrix of the lines is proved by hand and checked by computer, not formalized. The Lean certificate is `LEAN_CERTIFICATE_WATERMARK_v2`; the certificate, the Lean project and all the evidence are in the repository github.com/tretoef-estrella/watermark-theorem.

Abstract

Let $S_m \subset \mathbb{P}^3$ be the complex Fermat surface $x_0^m + x_1^m + x_2^m + x_3^m = 0$ of prime degree $m \geq 5$, and let V be the sublattice of $H^2(S_m, \mathbb{Z})$ generated by the classes of its $3m^2$ lines. We prove the closed formula

$$|\text{disc } V| = m^{3(m-3)^2}.$$

Since the lines generate the full Néron–Severi group for $\gcd(m, 6) = 1$ (Schütt–Shioda–van Luijk; Degtyarev), this computes $|\text{disc NS}(S_m)|$ exactly for every prime $m \geq 5$ and answers, for prime degree, the determinant question raised by Shioda in 1987 (Questions 7.2/7.4 of [7]). The proof is a chain of elementary, fully explicit lattice computations: a pairing lemma by character orthogonality, a trace-lattice discriminant by a Ramanujan-sum circulant, a transport identity, one exactly evaluated determinant, a blockwise saturation count, a kernel-saturation theorem ($mK \subseteq \text{SAT}$), and a 12-dimensional relation count over F_m solved in closed form. Every lemma was verified independently in a second computational environment, and the global identity was confirmed numerically at $m = 5, 7$ and, through a sealed pre-registered prediction, at $m = 11$.

The companion note [B] derives the corollaries: the discriminant of the transcendental lattice and an explicit closed recipe for the Brauer group order of the reductions of S_m over finite fields.

1. Introduction

The Néron–Severi lattice of the Fermat surface has been a touchstone of the theory of algebraic cycles since Shioda’s work of 1979–1987. For degree m with $\gcd(m, 6) = 1$, the $3m^2$ lines on S_m generate $\text{NS}(S_m)$ integrally — this is the theorem of Schütt–Shioda–van Luijk [4] completed by Degtyarev [5], resolving the question posed by Aoki–Shioda [3]. The *rank* of $\text{NS}(S_m)$ has been known in closed form since [3]:

$$\text{rank } \text{NS}(S_m) = 3(m-1)(m-2) + 1.$$

The *discriminant*, by contrast, resisted closure. Shioda asked for its structure in 1987 ([7], p. 133, Questions 7.2 and 7.4); Schütt–Shioda–van Luijk verified the value $|\text{disc NS}| = m^{3(m-3)^2}$ computationally for odd $m \leq 81$ ([4], around eq. (16)); Aljovin–Movasati–Villaflor [6] computed the elementary divisors of these lattices cell by cell for degrees up to 14 (their Table 1 — the same rows our instruments reproduce byte-exact in §13), at hardware costs reaching 170 GB of swap memory for a single cell, while their largest cells (the linear-cycle lattices behind their Table 2) remained beyond computation. To our knowledge no proof of the closed formula, valid for all prime m , existed in the literature: each published value is an isolated machine verdict.

This note proves it. The argument is deliberately elementary: it never leaves the explicit integer Gram matrix of the line configuration, and every step is either a character-orthogonality computation, a circulant determinant, or a finite-dimensional count over F_m carried out in closed form. The structure of the proof — discriminants of explicit trace sublattices, one big determinant, and the saturation index of the Gram kernel — was discovered by measurement (the campaign’s instruments and run logs are part of the public record; see §13) and then closed by pen and paper. The same formula, with the extra factor $(m/4)^3$ for even m , fits all published data and seven sealed blind predictions of the campaign (the *Watermark Law*, [W]); only the odd-prime half is proven here, and the scope is stated precisely in §2.

2. The configuration, the lattice, and the statement

The surface and its lines. S_m : $x_0^m + x_1^m + x_2^m + x_3^m = 0$ in $P^3(C)$, $m \geq 5$ prime. S_m contains exactly $3m^2$ lines [4], organized in three families of m^2 according to the pairing of coordinates; in the standard parametrization (the one used in [4] and [6]) each family is indexed by $(k, l) \in (Z/m)^2$. By adjunction every line has self-intersection $2 - m$; two distinct lines of the same family meet (in one point) precisely when they share one of their two indices; lines of different families meet precisely when a single linear condition on the indices holds. In the coordinates fixed throughout this note (the campaign’s session coordinates, which realize the standard configuration):

- within a family: $L_{\{(k,l)\}} \cdot L_{\{(k',l')\}} = 1$ if $k = k'$ or $l = l'$ (and $(k,l) \neq (k',l')$), 0 otherwise; self-intersection $2 - m$;
- family 1 $\times$ family 2: incidence $\iff k - l \equiv k' - l' \pmod{m}$;
- family 2 $\times$ family 3: incidence $\iff k + l \equiv k' + l' \pmod{m}$;
- family 1 $\times$ family 3: incidence $\iff k' - l' \equiv k + l + 1 \pmod{m}$.

Let $A = \bigoplus_{j=1,2,3} \mathbb{Z}^{\{(\mathbb{Z}/m)^2\}}$ be the free abelian group on the lines and G the $3m^2 \times 3m^2$ intersection Gram above. Define

$V := A / \ker G$, the *line lattice*, with the induced bilinear form.

All results below are theorems about this explicitly defined integer matrix; the geometric identification of G with the intersection form of the $3m^2$ lines is the classical configuration of [4], [6].

Theorem (the Watermark Theorem, odd-prime half). *For every prime $m \geq 5$,*

$$|\text{disc } V| = m^{3(m-3)^2}.$$

Corollary 1 (Shioda's question, prime degree). *For prime $m \geq 5$, $\gcd(m, 6) = 1$, so $V = NS(S_m)$ by [4, Thm 1.1] and [5]; hence $|\text{disc } NS(S_m)| = m^{3(m-3)^2}$, answering Questions 7.2/7.4 of [7] for prime degree.*

Scope. $m = 3$ is degenerate for this machinery (the four-pencil argument of Lemma R.2 and the degree count of Lemma C require $m \geq 5$) and its disc exponent is 0. Odd composite m and the even half of the Watermark Law (the extra factor $(m/4)^3$) are **not claimed here**: they remain conjectural, supported by the measured evidence catalogued in [W] (all twelve published AMV rows and seven sealed blind predictions).

Notation. For $t \in \mathbb{Z}/m$ put $x_t(k, l) = k + t \cdot l$, and $x_\infty(k, l) = l$. The $m + 1$ **pencils** of a family are the level-set partitions of these forms ($t \in \{0, \infty\}$ the two **axis** pencils; the rest **live**). pb_L denotes pullback along x_L ; $Z_0 \subset \mathbb{Z}^m$ the sum-zero sublattice; N_j the all-ones vector of family j ; $\text{aug}(u) = \sum u(p)$; $c_m(x) = m \cdot [x \equiv 0] - 1$ the prime- m Ramanujan function; $f_r := m \cdot e_r - 1 \in Z_0$, so $c_m(x - r) = f_r(x)$. The translation torus $\Gamma \cong (\mathbb{Z}/m)^3$ acts on A by isometries; its characters act on the families through the dictionary $\text{fam1} \leftrightarrow (a, b, 0)$, $\text{fam2} \leftrightarrow (a, b, a+b)$, $\text{fam3} \leftrightarrow (a, a, a+b')$, and a family pencil corresponds to a punctured line of $(\mathbb{Z}/m)^2$ -characters. The **trace vectors** are $v_{\{j, (a,b), r\}} :=$ the vector of family j with coordinates $c_m(ak + bl - r)$.

3. Lemma P (pairings, dead axes, phases, census)

Write $D_x := \sum_{k \in \mathbb{Z}/m} \zeta^{xk} = m \cdot [x \equiv 0]$ and $w_j(a, b) := \sum_{k, l} \zeta^{ak+bl} L_j(k, l)$. Splitting each Gram block into its diagonal and incidence parts and summing geometric series gives closed pairing formulas; in particular:

(P1) For $a, b \neq 0$: $w_j(a, b) \cdot w_j(-a, -b) = (2-m)m^2 - m^2 - m^2 = -m^3$, uniformly in m . **(P2)** For axis characters $((a, 0)$ or $(0, b))$ the self-pairing vanishes and $w \perp H$ (the hyperplane class); negative-definiteness on the primitive part forces the axis classes to vanish in V — the *dead axes*. **(P-cross)** Blocks (1,2) and (2,3): $w_j(a, b) \cdot w_{j'}(a', b') = D_{\{a+a'\}} D_{\{a+b\}} D_{\{a'+b'\}}$. Block (1,3): the twisted incidence contributes one explicit phase: $w_1(a, b) \cdot w_3(a', b') = \zeta^{\{a'\}} \cdot D_{\{a+a'\}} D_{\{b+a'\}} D_{\{a'+b'\}}$. **(P3')** On the three overlap families of Γ -characters (where two family dictionaries collide) the supporting trace classes satisfy the null relations $v_{\{j, r\}} + v_{\{j', r+s_0\}} \in \ker G$, with phase law $s_0 = 0$ for the pairs (1,2), (2,3) and $s_0 = 1$ (the $\zeta^{\{a'\}}$ phase at function level) for (1,3). **(P4)** Inclusion-exclusion over the three blocks gives $3(m-1)^2 - 3(m-1) = 3(m-1)(m-2)$ live characters, each of multiplicity one in $V \otimes Q$ — reproving the rank formula of [3] from scratch.

Corollary 5 (kernel census). $\ker G \otimes Q =$ six axis character lines (rank $6(m-1)$) $\oplus$ three overlap null lines (rank $3(m-1)$) $\oplus$ a 2-dimensional trivial part; total rank $9m - 7$. In particular a kernel vector has zero isotypic mass on every live non-overlap pencil.

Verification: every displayed pairing checked exhaustively over all characters at $m = 5$ and 7 , including the (1,3) phase on all m^4 cases per prime, in two independent environments.

4. Lemma T (the trace-lattice discriminant)

For a live orbit O supported on family j with pencil form (a', b') , the $m - 1$ independent trace vectors v_r have Gram $B(v_r, v_s) = \sum_t \zeta^{-t(r-s)} (w(t\alpha) \cdot w(-t\alpha)) = -m^3 \cdot c_m(r-s)$ by (P1); the $(m-1) \times (m-1)$ matrix $-m^3(mI - J)$ has $|\det| = m^{3(m-1)} \cdot m^{m-2}$, since $\det(mI - J)_{m-1} = m^{m-2}$. Hence for every live orbit

$$\text{disc } T(O) = \pm m^{4m-5}.$$

5. The transport identity

Let $K := \ker G$ (saturated, rank $9m - 7$) and let $F \subset A$ be the Z -span of one trace lattice per live orbit together with the trivial lift N_1 . Then $F \cap K = 0$ and

$$\text{unified} := [V : \varphi(F)] = [A : F \oplus K] = |\det[\text{basis of } K \mid \text{generators of } F]|,$$

where $\varphi: A \rightarrow V$ is the quotient map. Stacking the explicit kernel spawn K_0 (the Lemma-P basis: axis trace systems, paired overlap null systems, $N_1 - N_2, N_2 - N_3$) instead of a kernel basis measures unified $\cdot [K : K_0]$; the trivial lift satisfies $\varphi(N_1)^2 = m^3 = m^2 \cdot e^2$ where e generates the rank-1 trivial piece ($e^2 = m$), so $\varphi(N_1) = \pm m \cdot e$ contributes one exact factor m to the bookkeeping.

6. The big determinant

$$|\det[K_0 \mid F \mid N_1]| = m^{\{(9m^2-3m-6)/2\}}, \text{ exactly over } Z.$$

Proof. Under the standard inner product on A : (i) different families have disjoint supports, hence are orthogonal; (ii) vectors built on different character lines within one family are orthogonal, because $\sum_{k,l} c_m(ak+bl-r) \cdot c_m(a'k+b'l-s) = (\sum_x c_m)(\sum_y c_m) = 0$ for independent forms — the map $(k,l) \mapsto (x,y)$ is a bijection of $(Z/m)^2$ and Ramanujan sums are sum-zero. The standard Gram of the full stack is therefore block diagonal, and every block is a circulant of the pattern $\mu(m\delta - 1)$ with $\det(mI - J)_{m-1} = m^{m-2}$: six axis blocks and $3m - 9$ live non-overlap blocks contribute m^{3m-4} each (to $\det^2$); the three overlap blocks have pattern $m^2(mI - J) \otimes [[1,1],[1,2]]$ with $\det[[1,1],[1,2]] = 1$, contributing m^{6m-8} each; the trivial 3×3 block on $\{N_1 - N_2, N_2 - N_3, N_1\}$ has $\langle N_i, N_j \rangle = m^2 \delta_{ij}$, contributing m^6 . Total exponent $6(3m-4) + (3m-9)(3m-4) + 3(6m-8) + 6 = 9m^2 - 3m - 6$, halved for the determinant. Every block determinant is $\pm$ a power of m , so the identity is exact over Z . ■

Verification: $v_m = 102$ ($m = 5$) and 207 ($m = 7$) with non- m part exactly 1, both environments.

7. Blockwise saturation

Let $\text{SAT} \subset K$ be the explicit lattice: the six axis pullback blocks $\text{pb}_{\{j,\text{axis}\}}(Z_0)$; the three paired overlap blocks $\{\text{pb}_j(y) + \text{pb}_{j'}(y \circ \tau_{s_0}) : y \in Z_0\}$ (τ = cyclic translation, s_0 the $P3'$ phase); and $Z(N_1 - N_2) \oplus Z(N_2 - N_3)$. Membership $\text{SAT} \subset K$ (which re-proves the phase law at the saturated level) and rank $9m - 7$ are file-verified. Each of the nine blocks saturates its trace sub-block by exactly m^{m-2} : the index $[Z_0 : \text{span}\{f_r\}]$ satis-

fies $\text{index}^2 = \text{disc}(\text{span } f_r) / \text{disc}(Z_0) = m^{2m-3}/m$, by the same circulant determinant; identically for the paired blocks, whose 2-power disc parts cancel in the ratio. Hence

$$[\text{SAT} : K_0] = m^{9(m-2)}.$$

8. Lemma R ($mK \subseteq \text{SAT}$)

R.1 (incidence identity). On each family, $\Sigma_{\{L \text{ over the } m+1 \text{ pencils}\}} E_L = m \cdot I + J$, where $(E_L u)(p) = \Sigma_{\{p' \in \text{fiber}_L(p)\}} u(p')$. *Proof:* a nonzero difference vector of $(Z/m)^2$ lies in exactly one of the $m+1$ directions; the diagonal is counted $m+1$ times. ■

R.2 (augmentation divisibility). For $x = (x_1, x_2, x_3) \in K$: $\text{aug}(x_1) + \text{aug}(x_2) + \text{aug}(x_3) = 0$ exactly, and $m \mid \text{aug}(x_j)$ for each j . *Proof:* $G \cdot N_j = m \cdot 1_A$ by direct incidence counting, giving the exact sum on pairing with x . For the divisibilities, let y be the indicator of the family-1 fiber $\{k + t \cdot l \equiv c\}$, $t \neq 0$. Within family 1, $(Gy)_1 = 2 \cdot 1_1 - m \cdot y$ (for $t \neq 0$ the fiber is a graph over k and over l). Across families the incidence condition restricted to the fiber is a bijection except at one exceptional slope per block: $G \cdot y = 2 \cdot 1_1 + 1_2 + 1_3 - m \cdot y$ for $t \notin \{0, 1, -1\}$; $= 2 \cdot 1_1 + 1_2 + m \cdot z_3 - m \cdot y$ at $t = 1$; $= 2 \cdot 1_1 + m \cdot z_2 + 1_3 - m \cdot y$ at $t = -1$ (z the displaced fiber indicator in the exceptional family). Such t exists since $m \geq 5$. Pairing with x and writing $a_j = \text{aug}(x_j)$: $2a_1 + a_2 + a_3 \equiv 0$, $2a_1 + a_2 \equiv 0$, $2a_1 + a_3 \equiv 0 \pmod{m}$; subtracting, $a_2 \equiv a_3 \equiv 0$ and $2a_1 \equiv 0$ with m odd. ■ (All displayed identities verified entrywise for every t, c , both primes.)

R.3 (overlap assembly). For $x \in K$ and an overlap pair $(L \text{ in family } j) \leftrightarrow (L' \text{ in family } j')$ with shift s_0 : the sum-zero parts of the fiber-sum vectors satisfy $\tilde{f}_{\{j', L'\}} = \tilde{f}_{\{j, L\}} \circ \tau_{\{s_0\}}$. *Proof:* for pencils $L'' \neq L$ the map $p \mapsto (x_L(p), x_{L''}(p))$ is a bijection, so L -fibers kill L'' -isotypic mass; the trivial component contributes only the mean; hence $\tilde{f}_{\{j, L\}} = m \cdot g$ with $\text{pb}_L(g)$ the L -isotypic component of x_j . By the kernel census the only live pencils carrying kernel mass are the overlaps, where the component lies in the one-dimensional null direction of P_3' ; multiplying the a -th Fourier coefficient by $\zeta^{a \cdot s_0}$ is translation by s_0 , so $g' = g \circ \tau_{\{s_0\}}$. The pair $\text{pb}_j(\tilde{f}) + \text{pb}_{j'}(\tilde{f} \circ \tau)$ is then a paired-block member of SAT. ■

R.4 (theorem: $mK \subseteq \text{SAT}$). For $x \in K$, by R.1: $m \cdot x_j = \Sigma_L \text{pb}(\tilde{f}_{\{j, L\}}) - \text{aug}(x_j) \cdot 1_j$; the fiber means are integers by R.2; writing $f = \tilde{f} + \text{mean}$: $m \cdot x_j = \Sigma_L \text{pb}(\tilde{f}_{\{j, L\}}) + (\text{aug}(x_j)/m) \cdot 1_j$. Axis terms lie in the axis blocks of SAT; live non-overlap sum-zero parts vanish by the census; the six overlap terms group into the three R.3 pairs, members of SAT; and the constant leftover $\Sigma_j (\text{aug}_j/m) \cdot 1_j$ has integer coefficients summing to zero (R.2), hence lies in the span of $N_1 - N_3, N_2 - N_3$. So $mK \subseteq \text{SAT}$. ■ (Independent fire test: $m \cdot x \in \text{SAT}$ verified exactly for all 38 kernel basis vectors at $m = 5$.)

R.5. K/SAT is elementary abelian of exponent m ; $[K : \text{SAT}] = m^d$ with $d = (9m-7) - \text{rank}_{\{F_m\}}(\text{SAT mod } m)$, and **the index has no part at any prime $p \neq m$** — by proof, not by measurement.

9. Lemma C (the counting lemma: $d = 12$)

$$\text{rank}_{\{F_m\}}(\text{SAT mod } m) = 9m - 19; \text{ equivalently } d = \dim_{\{F_m\}} K/\text{SAT} = 12, \text{ for every prime } m \geq 5.$$

Proof. Functions $F_m \rightarrow F_m$ are uniquely polynomials of degree $\leq m-1$, and $\Sigma_{\{x \in F_m\}} x^e = -1$ if $(m-1) \mid e$, $e > 0$, and 0 otherwise; hence the mod- m span of the sum-zero column functions is exactly the polynomials of degree $\leq m-2$. An F_m -relation among the SAT columns is a choice of polynomials $u_{\{j,k\}}$,

$u_{\{j,l\}}$ (axis blocks), w_{12}, w_{23}, w_{13} (overlap blocks; one polynomial feeds both supporting families, the (1,3) block through the translation τ), and scalars $\varepsilon_1, \varepsilon_2$ (the N-differences), satisfying three identities in $F_m[k,l]$ of degree $\leq m-2 < m$ (so the function-polynomial dictionary is faithful):

$$(F1) \ u_{\{1,k\}}(k) + u_{\{1,l\}}(l) + w_{12}(k-l) + w_{13}(k+l) + \varepsilon_1 = 0 \quad (F2) \ u_{\{2,k\}}(k) + u_{\{2,l\}}(l) + w_{12}(k-l) + w_{23}(k+l) + (\varepsilon_2 - \varepsilon_1) = 0 \quad (F3) \ u_{\{3,k\}}(k) + u_{\{3,l\}}(l) + w_{23}(k+l) + (w_{13} \circ \tau)(k-l) - \varepsilon_2 = 0.$$

Mixed monomials $k^{d-r} l^r$ ($1 \leq r \leq d-1$) arise only from the two live-pencil terms; in degree d their coefficient is $C(d,r) \cdot [b_d + (-1)^r a_d]$, and $C(d,r) \not\equiv 0 \pmod{m}$ for $d \leq m-2$. For $d \geq 3$ the cases $r = 1, 2$ force $a_d = b_d = 0$; for $d = 2$ the single case $r = 1$ forces equality of the two quadratic coefficients, and chaining (F1)–(F3) (translation preserves the leading coefficient) makes all three equal to one common q — whose realizability is the polarization identity $(k-l)^2 + (k+l)^2 = 2k^2 + 2l^2$. Once the mixed terms cancel, the remaining content of each identity is k -pure plus l -pure, determining each family's axis pair up to its constant ambiguity ($c, -c$); the determined parts have degree $\leq 2 \leq m-2$. The solution space is therefore parametrized freely by: the common quadratic q (1); the linear coefficients of the three w 's (3); their constants (3); $\varepsilon_1, \varepsilon_2$ (2); the three constant splits (3) — total **12**, for every prime $m \geq 5$. ■

Verification: direct rank measurements 26 and 44 ($= 9m - 19$) by two independent constructions, plus the twelve relations exhibited explicitly (file COUNTING_LEMMA_PROBE_v1.py) and checked valid of rank 12 at both primes.

$$\text{Hence } [K : K_0] = [K : \text{SAT}] \cdot [\text{SAT} : K_0] = m^{12} \cdot m^{9(m-2)} = m^{9m-6}.$$

10. Assembly: proof of the Theorem

By §§5–9, $\text{unified} = |\det| / (m \cdot [K : K_0]) = m^{(9m^2-3m-6)/2 - 1 - (9m-6)} = m^{(9m^2-21m+4)/2}$. By §4 and Lemma T,

$$\text{disc } V = \text{disc}(\oplus_{\mathcal{O}} T(\mathcal{O}) \oplus Z \cdot e) / \text{unified}^2 = (m^{4m-5})^{3(m-2)} \cdot m / m^{9m^2-21m+4} = m^{(12m^2-39m+30) + 1 - (9m^2-21m+4)} = m^{3m^2-18m+27} = m^{3(m-3)^2}. \blacksquare$$

(The orbit count is $3(m-2) = 3(m-1)(m-2)/\varphi(m)$ for prime m .)

11. Sign and small cases

V contains the ample hyperplane class, so its form has hyperbolic signature $(1, \text{rank } V - 1)$ and $\text{disc } V = (-1)^{\text{rank } V - 1} |\text{disc } V|$ with $\text{rank } V = 3(m-1)(m-2) + 1$. $m = 3$: the exponent $3(m-3)^2$ is 0 and $\text{disc } V = \pm 1$ (classical; outside this proof's machinery). Odd composite and even degrees: conjectural — see [W].

12. Corollaries beyond the statement

(i) **Shioda's Questions 7.2/7.4** ([7], p. 133) are answered for prime degree (Corollary 1 above). (ii) **The transcendental lattice** $T_m = \text{NS}^\perp \subset H^2(S_m, \mathbb{Z})$ has $|\text{disc } T_m| = m^{3(m-3)^2}$ and rank $(m-1)(m-3)^2$ — see the companion note [B], where this is derived from unimodularity of H^2 in five lines. (iii) **The Brauer group** of the reductions of S_m over finite fields acquires a closed-form order recipe through the Artin–Tate formula, every ingredient of which is now explicit — see [B]. (iv) The glue group $V/(\oplus \text{pieces})$ is proven elementary abelian of F_m -dimension $(3m^2-9m+4)/2$, a structure result on the integral cohomology of independent interest.

13. Computational verification and provenance

Every measured number in this note reproduces from instruments in the public repository: GRANSLAP_PROBE_v1/v2 (the big determinant, the independent full-kernel index, the SNF of G — whose nonzero elementary-divisor profile coincides with the published AMV rows), LEMAR_SHAKE_PROBE_v1 (residual anatomy at both primes), BETA1_DISPLAY_PROBE_v1 (the R.2 display identities for every parameter; the three pure generators), COUNTING_LEMMA_PROBE_v1 (the rank and the twelve explicit relations). The development followed a dual-instance protocol: every lemma was independently re-derived, and every computation independently reproduced, in a second environment before ratification; two intermediate conjectures were killed by their own pre-registered tests (the failures are documented in the campaign findings), and the $m = 11$ cell was decided by a sealed prediction written before computation — 27 orbit pieces, each of discriminant 11^{17} , glue 11^{134} — which passed on all counts (engine LACAZADELOSPELADOS, run log in repository; MacBook Air M2, 8 GB, single thread, 25% CPU). The final identities of §§6–9 imply, for the $m = 11$ rail, big determinant 11^{525} , $[K : K_0] = 11^{93}$, unified 11^{431} , all consistent with the sealed run.

References

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7. T. Shioda, *Some observations on Jacobi sums*. Advanced Studies in Pure Mathematics **12** (1987), 119–135.

[W] *The Watermark Law* (this repository: THE_WATERMARK_LAW.md) — the full empirical law, the twelve AMV rows, the seven sealed blind predictions, and the published off-by-one erratum. [B] *Corollaries of the Watermark Theorem: the transcendental lattice and the Brauer group* (this repository: THE_BRAUER_COROLLARY.md).

Numbers from files only. Every claim above is either proven in the text or anchored to a named instrument and run log in the repository.